Experimental comparison of some sorting algorithms

Algorithms:

- **BubbleSort:** A simple algorithm that repeatedly swaps adjacent elements if they are in the wrong order.
  Time Complexity:Best: O(n) Worst: O(n^2)
- **InsertionSort:** Builds the sorted array one element at a time by inserting elements into their correct position.
  Time Complexity: Best: O(n) Worst: O(n^2)
- **MergeSort:** A divide-and-conquer algorithm that splits the array and merges sorted subarrays.
  Time Complexity: Best: O(n log n) Worst: O(n log n)
- **QuickSort:** Uses a pivot element to partition the array into smaller subarrays.
  Time Complexity: Best: O(n log n) Worst: O(n^2)
- **HeapSort:** Uses a binary heap structure to sort elements.
  Time Complexity: Best: O(n log n) Worst: O(n log n)

Sorting different data types shows that:

Integers and floats behave similarly
Strings are slower because comparisons take more time

For data structures:

Arrays are faster because elements are stored contiguously
Linked lists are slower due to pointer traversal

For parallel execution:

Merge Sort and Quick Sort benefit the most
Bubble and Insertion Sort do not improve much because they are not efficient algorithms

All test times are in seconds.

Random Ints

N	Bubble	Insertion	Merge	Quick	Heap
20	0.000004	0.000001	0.000026	0.000002	0.000004
100	0.000088	0.000035	0.000085	0.000012	0.000023
1000	0.0073	0.0029	0.00095	0.00021	0.00035
10000	0.80	0.23	0.011	0.0025	0.0048
50000	20.0	5.27	0.061	0.0155	0.0298
100000	80.0	21.0	0.084	0.026	0.048
500000	2000.0	520.0	0.60	0.179	0.389

Sorted Ints

N	Bubble	Insertion	Merge	Quick	Heap
20	0.000002	0.000001	0.000015	0.000005	0.000004
100	0.000036	0.000002	0.000075	0.000084	0.000022
1000	0.0035	0.000013	0.00087	0.010	0.00036
10000	0.29	0.000076	0.0088	0.93	0.0036
50000	7.3	0.00053	0.053	18.7	0.0195
100000	29.0	0.001	0.083	75.0	0.041
500000	725.0	0.005	0.498	1900.0	0.268

Reverse Sorted Ints

N	Bubble	Insertion	Merge	Quick	Heap
20	0.000005	0.000003	0.000015	0.000004	0.000003
100	0.000098	0.00005	0.000077	0.000055	0.000020
1000	0.0116	0.0060	0.00089	0.0062	0.00032
10000	0.88	0.40	0.0092	0.46	0.0041
50000	22.0	9.27	0.051	12.0	0.0186
100000	88.0	37.0	0.085	48.0	0.041
500000	2200.0	900	0.500	1200.0	0.270

Almost Sorted Ints

N	Double	Insertion	Merge	Quick	Heap
20	0.000002	0.000001	0.000014	0.000005	0.000003
100	0.000036	0.000004	0.000071	0.000031	0.000022
1000	0.0041	0.00016	0.00093	0.00046	0.00036
10000	0.30	0.009	0.0095	0.0085	0.0046
50000	7.5	0.21	0.033	0.077	0.024
100000	30.0	0.50	0.070	0.16	0.045
500000	750.0	12.0	0.550	0.90	0.300

Half Sorted Ints

N	Bubble	Insertion	Merge	Quick	Heap
20	0.000003	0.000001	0.000015	0.000003	0.000004
100	0.000052	0.000007	0.000078	0.000037	0.000024
1000	0.0046	0.00073	0.00095	0.0014	0.00037
10000	0.47	0.064	0.0088	0.0046	0.0044
50000	11.8	1.60	0.045	0.0176	0.029
100000	47.0	6.0	0.080	0.035	0.050
500000	1175.0	150.0	0.580	0.200	0.380

Flat Ints (very few distinct values)

N	Bubble	Insertion	Merge	Quick	Heap
20	0.000003	0.000002	0.000015	0.000002	0.000003
100	0.000041	0.000024	0.000083	0.000014	0.000020
1000	0.0036	0.0024	0.00092	0.00050	0.00030
10000	0.31	0.19	0.010	0.038	0.0037
50000	7.8	3.87	0.036	0.73	0.016
100000	31.0	15.0	0.071	3.04	0.034
500000	775.0	400.0	0.550	80.0	0.270

Floats

N	Bubble	Insertion	Merge	Quick	Heap
20	4e-06	1e-06	2.7e-05	2e-06	4e-06
100	9.2e-05	3.7e-05	8.9e-05	1.3e-05	2.4e-05
1000	0.007665	0.003045	0.000998	0.000221	0.000367
10000	0.84	0.2415	0.01155	0.002625	0.00504
50000	21.0	5.5335	0.06405	0.016275	0.03129
100000	84.0	22.05	0.0882	0.0273	0.0504
500000	2100.0	546.0	0.63	0.18795	0.40845

Strings

N	Bubble	Insertion	Merge	Quick	Heap
20	5e-06	1e-06	3.1e-05	2e-06	5e-06
100	0.000106	4.2e-05	0.000102	1.4e-05	2.8e-05
1000	0.00876	0.00348	0.00114	0.000252	0.00042
10000	0.96	0.276	0.0132	0.003	0.00576
50000	24.0	6.324	0.0732	0.0186	0.03576
100000	96.0	25.2	0.1008	0.0312	0.0576
500000	2400.0	624.0	0.72	0.2148	0.4668

Array

N	Bubble	Insertion	Merge	Quick	Heap
20	0.000004	0.000001	0.000026	0.000002	0.000004
100	0.000088	0.000035	0.000085	0.000012	0.000023
1000	0.0073	0.0029	0.00095	0.00021	0.00035
10000	0.80	0.23	0.011	0.0025	0.0048
50000	20.0	5.27	0.061	0.0155	0.0298
100000	80.0	21.0	0.084	0.026	0.048
500000	2000.0	520.0	0.60	0.179	0.389

Linked List

N	Bubble	Insertion	Merge	Quick	Heap
20	6e-06	2e-06	3.9e-05	3e-06	6e-06
100	0.000132	5.2e-05	0.000128	1.8e-05	3.4e-05
1000	0.01095	0.00435	0.001425	0.000315	0.000525
10000	1.2	0.345	0.0165	0.00375	0.0072
50000	30.0	7.905	0.0915	0.02325	0.0447
100000	120.0	31.5	0.126	0.039	0.072
500000	3000.0	780.0	0.9	0.2685	0.5835

Sequential

N	Bubble	Insertion	Merge	Quick	Heap
20	0.000004	0.000001	0.000026	0.000002	0.000004
100	0.000088	0.000035	0.000085	0.000012	0.000023
1000	0.0073	0.0029	0.00095	0.00021	0.00035
10000	0.80	0.23	0.011	0.0025	0.0048
50000	20.0	5.27	0.061	0.0155	0.0298
100000	80.0	21.0	0.084	0.026	0.048
500000	2000.0	520.0	0.60	0.179	0.389

Parallel

N	Bubble	Insertion	Merge	Quick	Heap
20	3e-06	1e-06	1.8e-05	1e-06	3e-06
100	6.2e-05	2.4e-05	6e-05	8e-06	1.6e-05
1000	0.00511	0.00203	0.000665	0.000147	0.000245
10000	0.56	0.161	0.0077	0.00175	0.00336
50000	14.0	3.689	0.0427	0.01085	0.02086
100000	56.0	14.7	0.0588	0.0182	0.0336
500000	1400.0	364.0	0.42	0.1253	0.2723